

Kevin Saephanh

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Technical Skills

Languages: Java, Python, Go, JavaScript/TypeScript

Backend & Cloud: Spring Boot, FastAPI, Node.js, AWS, Docker, Kafka, Redis, OpenTelemetry

Frontend: React, Angular

Databases: PostgreSQL

Experience

Senior Software Engineer, Capital One July 2025 – Present

- Optimized **storage I/O performance** by delivering a **400x PostgreSQL query speedup (300s to 700ms)** on an Aurora backend via **zero-downtime covering-index migrations**. Eliminated **100% of per-query disk I/O wait**, drastically reducing system latency, resolving production 504s that drove a **25x vCPU oversubscription**.
- Engineered a **multithreaded**, precomputed caching system running **4,800 queries** across Spring Boot microservices, capping concurrency at 5 threads to minimize database connection pools, cutting data retrieval latency from **60s+ to under 300ms**, and resolving a related **Redis** race condition with **ShedLock**.
- Engineered core Go service enhancements and an **OpenTelemetry/New Relic** observability pipeline for a partner team on ECS Fargate. Designed selective traces across **100M+ SQS messages** to establish system visibility while **preventing mass ingestion overhead**.
- Built an enterprise-wide **AI-powered documentation engine** that auto-generates API references, schemas, and setup guides from Git repositories, with **intelligent change detection** to keep docs in sync across teams.

Software Engineer III, JPMorgan Chase July 2024 – July 2025

- Designed a Kafka and Oracle **distributed logging architecture**, aggregating telemetry across **10 microservices**, enabling rapid root cause identification of cascading failures and out-of-memory (OOM) errors during production incidents.
- Optimized **data processing throughput** for batch jobs handling **10M+ daily trade records**, cutting **execution time by 20%** by consolidating processing into a single microservice and adding **async Kafka triggers**.
- Implemented a **consistent-hashing strategy** for a tax processing Kafka consumer cluster, balancing node load and reducing processing **latency by 15%**.

Backend Engineer (Contract), Great Lakes Reality Labs Aug 2022 – Feb 2024

- Rebuilt a non compiling MVP backend from scratch in **NestJS** and **PostgreSQL**, implementing **context dependent RBAC** across three user types using role guards and ownership/association scoped queries.
- Developed a job matching engine to efficiently pair students with company roles based on **aggregate skills and eligibility rules**.
- Integrated a backend subscription system, launching the platform's **first revenue-generating feature**.

Full Stack Engineer, Cognizant Jan 2020 – Jan 2024

- Created a batch job to correct **30,000 data inconsistencies**, unblocking a backlog of enterprise auto loan investigations and audits.
- Migrated a BFF Spring Boot app behind AWS API Gateway, leveraging its built-in JWT authentication and **limiting ingress traffic** for downstream microservices.

Certifications

AWS Certified Solutions Architect - Associate Jan 2024 – Jan 2027

Education

Georgia Institute of Technology – M.S. in Computer Science (AI Specialization) Expected 2027
University of California, Santa Barbara – B.A. in Biological Anthropology 2019